

# MUHAMMAD BILAL SALEEM Full Stack & AI Engineer

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**Portfolio:** <https://bilal-saleem-portfolio.vercel.app/>

## SUMMARY

Result-oriented Computer Science undergraduate specializing in full-stack web development and AI-driven automation. Proficient in designing scalable backend architectures, robust frontend interfaces, and implementing secure authentication pipelines. Adept at leveraging modern frameworks to engineer clean, maintainable, and highly efficient software solutions.

## TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL (PostgreSQL)
- **Frontend Technologies:** React, Next.js, Redux, HTML5/CSS3, Bootstrap
- **Backend Frameworks:** Node.js, Nest.js, Express.js, FAST APIs
- **Databases & Cloud:** MongoDB, PostgreSQL
- **AI & Automation:** Agentic AI, LangGraph, RAG, OpenAI APIs, Scikit-learn, TensorFlow, NumPy, Pandas
- **Developer Tools:** Git, Postman, Docker, JWT, Multi-Provider OAuth 2.0, Stripe

## WORK EXPERIENCE

### Intern - CitrusBits July 2025 - Aug 2025

- **Architected scalable RESTful APIs** utilizing Nest.js and PostgreSQL to support core backend service features.
- **Engineered predictive data models** using Scikit-learn and TensorFlow, integrating machine learning pipelines with backend endpoints.
- **Optimized database queries** and schema designs in PostgreSQL, improving data retrieval efficiency during development cycles.

### Operations Manager - IEEE CS CUI Nov 2024 - Jan 2026

- **Directed execution operations** for COMPETE'24 and COMSATS Tech Summit (CTS), national-level hackathons attracting 500+ participants.
- **Coordinated cross-functional** teams handling logistics, technical platform setup, and sponsor relations to ensure seamless event delivery.

## EDUCATION

### Bachelor of Computer Sciences 2023 - 2026

- Comsats University Islamabad

## PROJECTS

### Clinical Appointment System (Nest.js, PostgreSQL, TypeScript, OAuth) | [Code Base](#)

- **Designed a secure backend system** handling clinical scheduling, patient records, and doctor workflows.
- **Integrated Google Calendar API** via OAuth 2.0 to automate real-time appointment syncing and minimize schedule conflicts.
- **Implemented automated email** notification triggers using Node.js backend queues to streamline patient communication.

### Car Dealership Platform (React, Node.js, Express.js, MongoDB) | [Go Live](#)

- **Built a full-stack e-commerce marketplace** allowing users to securely list, browse, and purchase vehicles via verified agent tiers.
- **Developed robust user authentication** using JWT and built scalable CRUD operations for dynamic vehicle inventory management.

### Resorts Web Application (Node.js, Express.js, MongoDB, JavaScript, Bootstrap) | [Go Live](#)

- **Architected a responsive full-stack listing platform** enabling users to seamlessly browse dynamic resort properties, read detailed reviews, and submit verified user feedback.

- **Integrated third-party APIs** to fetch and render real-time external data, enriching the user experience with accurate localization or pricing details.
- **Designed a structured MongoDB database schema** to efficiently manage relational-like data models for user accounts, reviews, and resort inventory.

#### **Agentic AI Knowledge Chatbot** (LangGraph, Python, RAG, OpenAI, Streamlit) | [Code Base](#)

- **Architected an autonomous AI Agent** using LangGraph to manage complex, multi-turn conversations with robust state management and fallback routing.
- **Implemented a Retrieval-Augmented Generation (RAG) pipeline** to anchor LLM responses in custom vector data, minimizing hallucinations.
- **Engineered dynamic tool-calling systems**, empowering the agent to programmatically decide when to call local internal utilities versus external APIs.

#### **Real-Time Collaborative Chat Application** (React, Express.js, MongoDB, Google OAuth 2.0) | [Code Base](#)

- **Developed a secure chat application** supporting standard email/password authentication alongside seamless Google OAuth 2.0 registration.
- **Designed efficient MongoDB schemas** to index, store, and seamlessly fetch high-throughput historical chat messages and room metadata.
- **Built responsive React interface modules** to synchronize state adjustments instantaneously during heavy, multi-user messaging traffic.